Ng Yu Hang (Mervin)

AI / ML Engineer | Generative AI | LLM Applications
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SUMMARY

AI engineer focused on generative AI and applied ML. I design and ship production LLM systems end to end: RAG pipelines, agents and tool use, evaluation and red-teaming, fine-tuning, and the APIs, interfaces and cloud deployments around them. I have delivered RAG and LLM-evaluation systems for real users at Singapore Polytechnic and in industry, with measured impact.

EXPERIENCE

Full-Stack AI Engineer (Intern), *SP Data Science & Analytics Centre (DSAC)* Mar 2026 - Present

- Built an Azure-based RAG chatbot for EM Services, a town council managing agent, cutting document retrieval time by ~70-85% and manual search time by ~60%.
- Built an LLM grading system for Power BI dashboards that reads the .pbix file, maps each visual to its student and grades against the rubric with cited evidence, reducing grading time by ~50-70%.
- Proposed and prototyped a project portal that matches DSAC's client projects with lecturers across SP's schools.

Product & AI (Part-time), *Influencees* Sep 2026 - Present

- Building AI features and product prototypes for a Singapore creator-marketing platform, including an internal business-development agent workspace and a creator Trust Check.

AI Researcher / Developer (Part-time), *PetaniAI* May 2026 - Aug 2026

- Evaluated new AI models and tools and turned them into working prototypes, including an interactive product site and a TikTok views leaderboard.

Student Developer, *Singapore Polytechnic, School of Computing* Sep 2025 - Feb 2026

- Built and evaluated an end-to-end generative-AI chatbot for a paid internal client project, using cloud services and both qualitative and quantitative user testing to iterate.
- Presented the piloted prototype to SP's Deputy Principal and PCEO, and showcased it to hundreds of SP staff at a seminar.

SELECTED PROJECTS

AgentLabs *(LLM evaluation, red-teaming, LoRA, FastAPI)*: Adversarial hardening for AI agents: attacks with 200 prompt-injection and tool-abuse cases, detects failures with deterministic verifiers, auto-patches prompts, tool policies and output filters, and retests; also fine-tunes small open models with LoRA. Live demo.

ProofBench *(FastAPI, PostgreSQL, Docker, CI)*: Evaluates software tools by generating adapters, running them in disposable sandboxes and scoring the output with a deterministic evaluator.

Lactic *(Graph RAG, embeddings, agents, Flutter)*: AI tutor app for Singapore A-Level students: hybrid vector and graph retrieval over 30k+ chunks, a deterministic agent harness and model routing through OpenRouter.

ESG Evidence Engine *(FastAPI, LLM agents, Sentinel-2)*: Research agent that checks companies' ESG claims against sourced evidence and satellite imagery, producing cited reports (PolyFintech 2026, team).

Edge Human Detection *(Computer vision, YOLO, PyTorch)*: Fine-tuned YOLO11 on thermal imagery for night-vision CCTV, with temporal filtering to cut false alarms and on-device inference.

HACKATHONS

PolyFintech 2026: ESG Evidence Engine (team) · **Edu2030 Vibe Hackathon, MOE HQ**: AI-literacy platform with an LLM prompt judge · **'Sup Build2026**: support-agent builder in 12 hours · **Ralphton**: one-shot autonomous app build · **OpenClaw Hackathon**: news-intelligence agent for GrowthLab

SKILLS

GenAI / LLM	RAG, embeddings and vector search, agents and tool use, LLM evaluation and red-teaming, prompt engineering, vision-language models, LoRA fine-tuning, OpenAI / OpenRouter / DeepSeek APIs, LangChain
ML	PyTorch, TensorFlow, Hugging Face Transformers / PEFT, scikit-learn, CNNs, YOLO, OpenCV, time-series forecasting
Languages	Python, TypeScript, JavaScript, SQL, Dart, Go
Backend & apps	FastAPI, Node.js, React, Next.js, Flutter
Deploy & data	Docker, Azure, Railway, GitHub Actions CI, PostgreSQL, Supabase, Power BI, Tableau

EDUCATION

Diploma in Applied AI & Analytics, *Singapore Polytechnic* 2024 - 2027
CGPA 3.83 / 4.00 · A / Distinction (top 5%) in AI modules including Deep Learning and AI & Machine Learning

GCE O-Level, *Chung Cheng High School (Main)* 2022 - 2023

CERTIFICATIONS & AFFILIATIONS

Certified AI Trainer, AI Singapore (selected for the Youth AI Train-the-Trainer Programme) · NVIDIA Fundamentals of Deep Learning · AWS Academy Cloud Foundations · Google Cloud skill badges (BigQuery data analytics and ML) · IBM AI Fundamentals · Member, Singapore Computer Society · Languages: English, Chinese